# NR-ReX Network Integration Project


*Presented by:*

- ✓ **Kavish** Gunasekara (Project Manager)
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- ✓ **Sarik** Ruwali (Business Analyst)


- 22 October 2025
- NIT1203 – Introduction to Project Management
- Group 03
- Victoria University Melbourne

# Introduction

INTEGRATION OF REX DEPOTS INTO NR'S CENTRAL SYSTEM

IMPROVED DATA VISIBILITY AND OPERATIONAL COORDINATION

WATERFALL MODEL WITH STRUCTURED CHECKPOINTS

20-MONTH PROJECT DURATION

# Project Overview

**Scope:**

- Integrate all ReX depots with NR’s central system
- Improve scheduling, tracking, and data reporting
- Provide full training and documentation

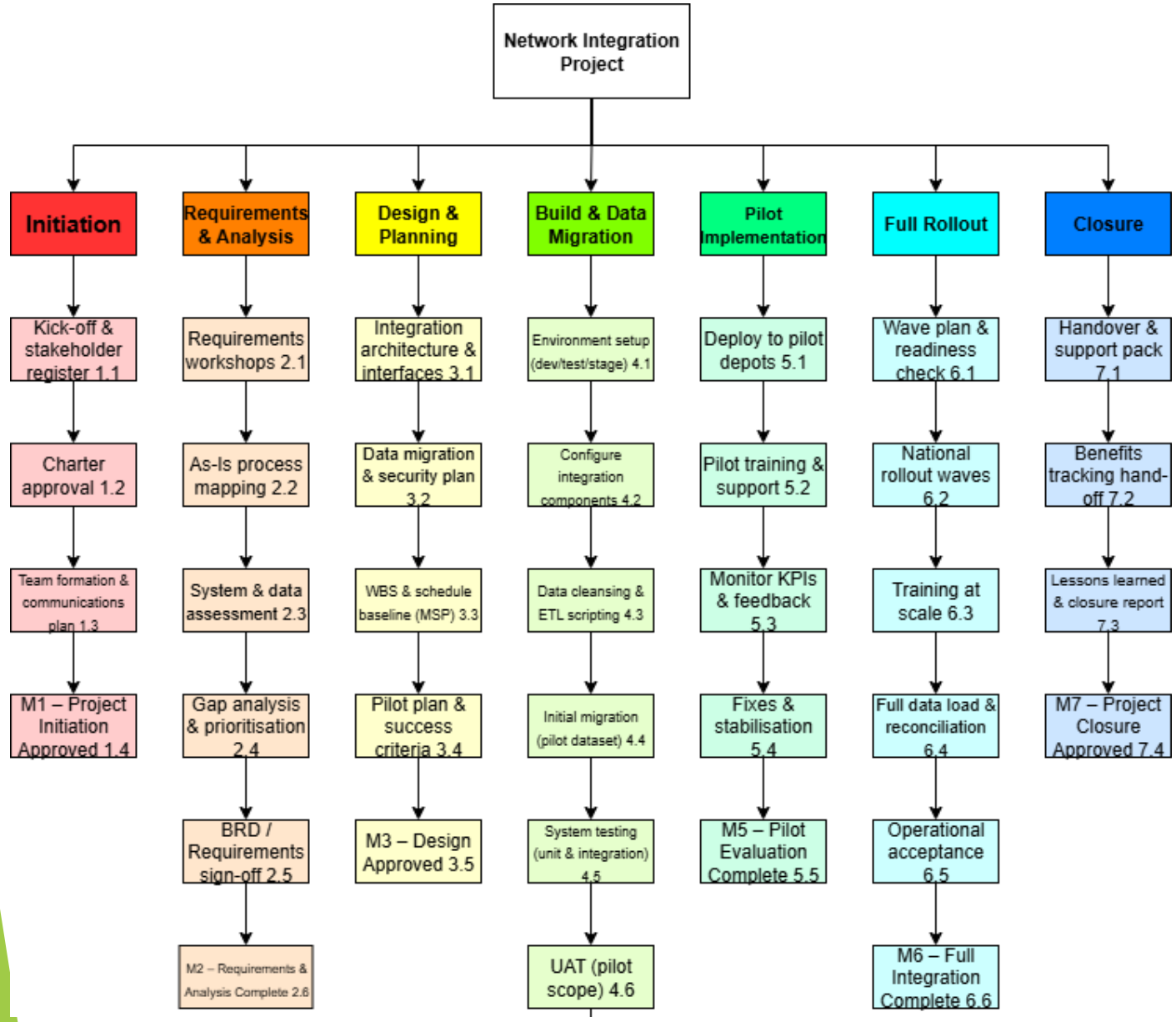
**Key Deliverables:**

- Integration architecture
- Data migration and security plan
- Pilot implementation report
- Final rollout documentation

| Stage | Duration | Focus | Milestone |
|---|---|---|---|
| Initiation | Months 1–4 | Requirements & Charter | M1 – Charter Approved |
| Design | Months 5–6 | System architecture | M2 – Design Approved |
| Build | Months 7–12 | Development & Migration | M3 – Build Completed |
| Pilot | Months 13–15 | Testing & Evaluation | M4 – Pilot Complete |
| Rollout | Months 16–19 | Full Deployment | M5 – Integration Complete |
| Closure | Month 20 | Final Sign-off | M6 – Project Closed |

# Project Timeline

# Work Breakdown Structure

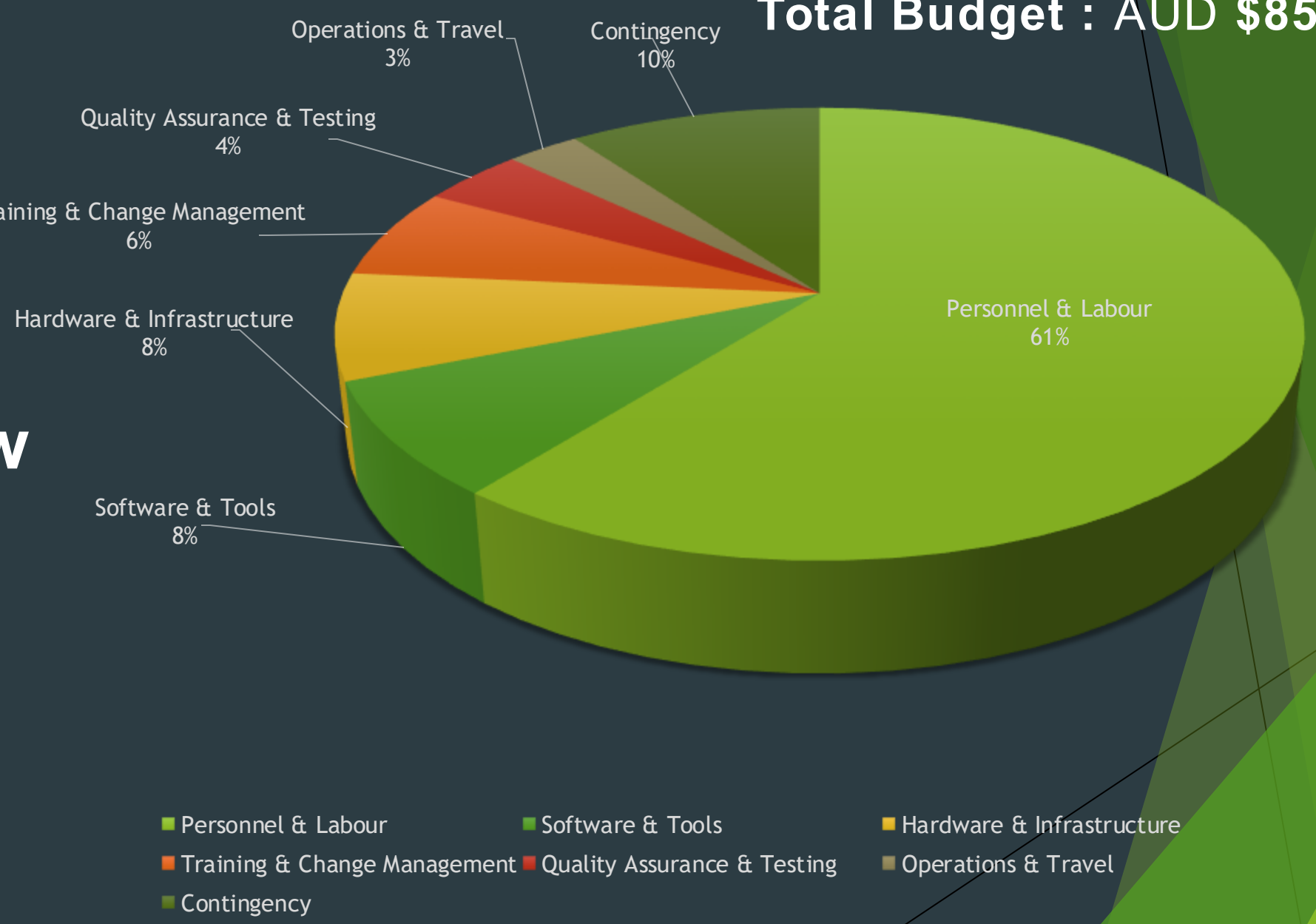


| Role | Name/ Group | Key Responsibilities |
|---|---|---|
| Project Manager | Kavish Gunasekara | Oversees planning, execution, communication, and ensures delivery of all project milestones. |
| Business Analyst | Sarik Ruwali | Gathers and documents requirements, maps business processes, supports UAT and change documentation. |
| Technical Lead | Darren Lim | Leads technical design, system architecture, integration setup, and testing coordination. |
| Change Manager | NR Staff Respresentative | Drives staff engagement, manages training, communication, and adoption of the new system. |
| Integration Engineers | ReX-NR IT Team | Handle API configuration, data migration, and network integration tasks. |
| Data Engineers | NR Data Support Team | Perform data cleansing, validation, and migration monitoring during build and rollout phases. |
| QA & Testing Lead | QA Consultant / Team | Conducts functional, system, and user acceptance testing; manages bug tracking and validation. |
| Trainers/ Support Staff | NR Training Department | Deliver pilot and rollout training sessions and provide post-implementation support. |
| Executive Sponsor | Dr. Ochini (VU) | Provides academic oversight, milestone approvals, and project evaluation. |
| Business Sponsor | NR Chief Operating Officer | Ensures project outcomes align with National Railways' strategic and financial goals. |

# Project Team

# Budget Overview

**Total Budget : AUD $850,000**



# NR & ReX Network Integration Project

| Project Budget Distribution | | Budget Summary |
|---|---|---|
| | | Budget (AUD) |
| 01/01/2025 – 31/08/2026 (20 months) | | $850,000.00 |
| **Tasks** | **Unit/ Description** | **Estimated Cost** |
| Personnel & Labour | | $515,000.00 |
| Project Manager | 20 months | $120,000.00 |
| Business Analyst | 18 months | $100,000.00 |
| Technical Lead | 18 months | $110,000.00 |
| Data Engineers | 2 persons *12 months | $120,000.00 |
| Trainers / Change Manager | Training design & delivery (pilot + rollout) | $35,000.00 |
| Quality Assurance & Testing Support | Contract / part-time | $20,000.00 |
| Administrative / Support Staff | Shared role | $10,000.00 |
| Software, Tools & Licensing | | $70,000.00 |
| Microsoft Project (planning & scheduling) | 3 licences | $4,500.00 |
| Jira & SharePoint (issue tracking & collaboration) | Annual subscription | $6,500.00 |
| Data Migration & Integration Tools (API, ETL) | Package licence | $18,000.00 |
| Security & Monitoring Tools (pen-test, logs) | Software package | $10,000.00 |
| Backup & Cloud Storage | Annual plan | $9,000.00 |
| QA & UAT Testing Tools | Pilot & rollout phase | $7,000.00 |
| Licence Renewal / Maintenance Fees | Annual contract | $5,000.00 |
| Hardware & Infrastructure | | $65,000.00 |
| Server Hosting & Configuration | 2 × medium servers | $15,000.00 |
| Network Equipment (routers, firewalls, VPN) | Implementation cost | $9,000.00 |
| Workstations for Integration & Testing | 5 × computers | $11,000.00 |
| Backup Devices & External Storage | 3 × devices | $7,000.00 |
| IT Security Hardware (encryption modules) | Set of devices | $8,000.00 |
| Hardware Maintenance & Upgrades | Ongoing | $10,000.00 |
| Training & Change Management | | $55,000.00 |
| Training Workshops & Manuals | 8–10 sessions | $18,000.00 |
| Online Learning & Certification Access | For NR & ReX staff | $9,000.00 |
| Pilot Training Support | On-site sessions | $13,000.00 |
| Staff Incentives & Engagement Materials | Communication campaigns | $5,000.00 |
| Travel & Lodging (trainer travel) | Limited sessions | $5,000.00 |
| Post-rollout Refreshers | 3 months after go-live | $5,000.00 |
| Testing & Quality Assurance | | $35,000.00 |
| Functional & Integration Testing | Build & pilot | $12,000.00 |
| Data Validation & UAT | Pilot & rollout | $9,000.00 |
| Security / Penetration Testing | Pre-rollout | $7,000.00 |
| Performance / Load Testing | Full rollout | $7,000.00 |
| Operations & Travel | | $25,000.00 |
| Project Meetings & Coordination | 20 months | $8,000.00 |
| Travel for On-site Work & Evaluation | 6–8 trips | $8,000.00 |
| Printing, Communication, Documentation | Materials & logistics | $4,000.00 |
| Stakeholder Workshops & Review Meetings | 3–4 sessions | $5,000.00 |
| Contingency Reserve (10%) | | $85,000.00 |
| Risk-related or unplanned costs (vendor delays, rework, tool upgrades, etc.) | Lump sum reserve | $85,000.00 |

# Detail Budget Plan

| Risk | Impact | Mitigation |
|---|---|---|
| Vendor delay | High | Contract-based delivery milestones |
| Data migration error | High | Pre-validation & backup recovery plan |
| Staff resistance | Medium | Change workshops & training |
| Integration failure | High | Testing at multiple pilot levels |
| Budget overrun | Medium | Monthly financial review |

# Risk Management

# Progress and Metrics

## Key Achievements

- Depot integration completed
- Faster operational turnaround
- System uptime
- Staff trained
- On-time, within budget

| Challenge | Solution |
| --- | --- |
| Staff reluctance | Communication workshops & incentives |
| Limited legacy data quality | Data cleansing & validation tools |
| Time pressure | Weekly progress checkpoints |
| Vendor dependency | Early contracting & follow-ups |

# Challenges and Solutions

# Conclusion

Delivered on time, within budget

All objectives and deliverables achieved

Improved efficiency for NR-ReX

Strengthened stakeholder collaboration

Continuous system monitoring

Quarterly review meetings

Ongoing staff training

Explore AI automation for data analytics

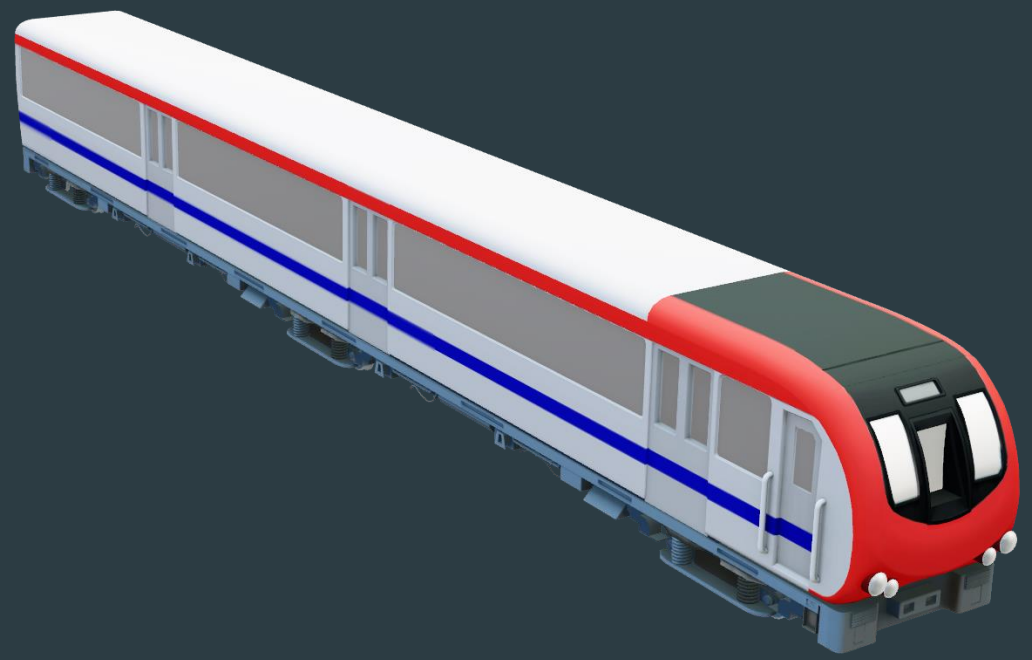
# Next Steps

# References

Victoria University. (2024). *Presentation tips & template (NIT1203)* [University guide (PDF document)]. Victoria University.

Schwalbe, Kathy. Information Technology Project Management (9th Edition), Cengage, 2018.

# Q&A



“Thank you for your attention.”
Questions are welcome..